

7 Large Sample Theory

7.1 Introduction

Large sample theory, also referred to as **asymptotic theory**, is a fundamental area of statistical inference that examines the behavior of estimators and test statistics as the sample size n becomes very large, i.e., $n \rightarrow \infty$. Rather than focusing on exact distributions for a fixed sample size, asymptotic theory studies the *limiting behavior* of statistical procedures.

In many practical situations, the exact finite-sample distributions of estimators or test statistics are either mathematically intractable or completely unknown, especially for complex models. Large sample theory provides a powerful alternative by offering **approximate results** that become increasingly accurate as the sample size grows.

Key Idea: As the sample size increases, random fluctuations due to sampling variability tend to average out, and many estimators exhibit regular and predictable behavior. This allows statisticians to approximate their distributions using well-known limiting distributions such as the normal or chi-square distributions.

Motivation for Large Sample Theory:

- Exact finite-sample results are often difficult to derive for complicated estimators.
- Many modern statistical models (e.g., regression, machine learning models) do not have simple closed-form distributions.
- Large datasets are increasingly common in practice, making asymptotic approximations highly relevant.

Core Principle: Large sample theory relies heavily on probability limit theorems such as:

- The Law of Large Numbers (LLN)
- The Central Limit Theorem (CLT)

These results ensure that estimators converge to true parameter values and often become approximately normally distributed.

Illustrative Insight: Consider the sample mean $\bar{X}$. Even if the underlying population is not normally distributed, the Central Limit Theorem guarantees that for sufficiently large n :

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

This approximation forms the basis for constructing confidence intervals and hypothesis tests in large samples.

Importance in Statistical Inference:

- Enables approximate confidence intervals and hypothesis tests.
- Provides theoretical justification for widely used estimation methods such as Maximum Likelihood Estimation (MLE).
- Forms the backbone of modern econometrics and data science methods.

Limitations:

- Results are approximate and may be inaccurate for small samples.
- The phrase “large sample” is context-dependent; what is large enough varies across applications.
- Some estimators may converge slowly, requiring very large samples for good approximations.

7.2 Finite-Sample vs Asymptotic Inference

Statistical inference can be broadly classified into two frameworks: **finite-sample inference** and **asymptotic (large-sample) inference**. The key distinction lies in whether conclusions are drawn using the *exact distribution* of a statistic for a fixed sample size or using its *limiting distribution* as the sample size becomes large.

7.2.1 Finite-Sample Inference

Finite-sample inference is based on the exact probability distribution of estimators or test statistics for a given sample size n . All conclusions (confidence intervals, hypothesis tests, p -values) are derived without approximation.

Key Characteristics:

- **Exactness:** Results are mathematically exact for the specified sample size.
- **Distribution Dependence:** Requires full knowledge of the sampling distribution.
- **Strong Assumptions:** Often relies on assumptions such as normality, independence, and identical distribution.
- **Limited Applicability:** Becomes difficult or impossible for complex models or estimators.

Common Distributions Used:

- Student's t -distribution (small-sample mean inference)
- Chi-square (χ^2) distribution (variance estimation)
- F -distribution (ANOVA and regression testing)

Example 1 (Sample Mean under Normality):

Let $X_1, X_2, \dots, X_n \sim \text{i.i.d. } N(\mu, \sigma^2)$. Then the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

has the exact distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Example 2 (Unknown Variance):

If σ^2 is unknown, the standardized statistic:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

follows a Student's t -distribution with $(n - 1)$ degrees of freedom:

$$T \sim t_{n-1}$$

Interpretation: Finite-sample inference is preferred when sample sizes are small because it provides exact results. However, its applicability is limited when the underlying distribution is unknown or when dealing with complex estimators.

7.2.2 Asymptotic Inference

Asymptotic inference studies the behavior of estimators and test statistics as the sample size $n \rightarrow \infty$. Instead of exact distributions, it relies on **limiting (approximate) distributions**.

Key Characteristics:

- **Approximate Results:** Valid when the sample size is sufficiently large.
- **Distribution-Free Flexibility:** Often does not require knowledge of the exact underlying distribution.
- **Reliance on Limit Theorems:** Based on results such as the Law of Large Numbers and Central Limit Theorem.
- **Wide Applicability:** Useful for complex models (e.g., MLE, regression, machine learning estimators).

Example 1 (Central Limit Theorem):

Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then:

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Equivalently,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n$$

Example 2 (Asymptotic Normality of Estimators):

Many estimators $\hat{\theta}$ satisfy:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$$

This allows construction of approximate confidence intervals:

$$\hat{\theta} \pm z_{\alpha/2} \sqrt{\frac{V}{n}}$$

Interpretation: Even when the exact distribution is unknown, asymptotic theory guarantees that estimators behave *approximately normally* for large samples, enabling inference using standard tools.

7.2.3 Comparison of Finite-Sample and Asymptotic Inference

Feature	Finite-Sample Inference	Asymptotic Inference
Nature of Results	Exact	Approximate
Sample Size Requirement	Small or fixed n	Large n
Distribution Knowledge	Required	Not always required
Assumptions	Strong (e.g., normality)	Often weaker
Applicability	Limited to simple models	Applicable to complex models
Accuracy	Exact	Improves as n increases

7.3 Asymptotic Concepts in Estimation

7.3.1 Consistency

An estimator $\hat{\theta}_n$ is **consistent** for parameter θ if:

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty$$

Types of Consistency:

- Weak consistency: convergence in probability
- Strong consistency: almost sure convergence

Example:

$$\bar{X} \xrightarrow{p} \mu$$

7.3.2 Asymptotic Unbiasedness

An estimator is asymptotically unbiased if:

$$\lim_{n \rightarrow \infty} E(\hat{\theta}_n) = \theta$$

7.3.3 Asymptotic Normality

An estimator $\hat{\theta}_n$ is asymptotically normal if:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$$

Interpretation:

- For large n , $\hat{\theta}_n$ is approximately normally distributed.
- Enables construction of confidence intervals.

7.3.4 Rate of Convergence

The **rate of convergence** describes how quickly an estimator $\hat{\theta}_n$ approaches the true parameter value θ as the sample size n increases.

Formal Definition:

An estimator $\hat{\theta}_n$ is said to converge to θ at rate a_n if:

$$a_n(\hat{\theta}_n - \theta) = O_p(1)$$

a_n is a deterministic sequence of positive numbers (i.e., it depends only on the sample size n , not on the data).

Equivalently,

$$\hat{\theta}_n - \theta = O_p\left(\frac{1}{a_n}\right)$$

Common Case ($\sqrt{n}$ -rate):

Most standard estimators satisfy:

$$\hat{\theta}_n - \theta = O_p\left(\frac{1}{\sqrt{n}}\right)$$

This is called the **$\sqrt{n}$ -rate of convergence**.

Interpretation:

- As n increases, the estimation error shrinks at the rate $1/\sqrt{n}$.
- Doubling the sample size reduces the error by approximately $1/\sqrt{2}$.
- Faster convergence implies more precise estimators.

Example 7.3.1

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with:

$$E(X_i) = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

Define the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Step 1: Mean and Variance of $\bar{X}$

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

This shows that the variability of $\bar{X}$ decreases as n increases.

Step 2: Apply the Central Limit Theorem (CLT)

The Central Limit Theorem states that:

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

This means that for large n , the scaled error $\sqrt{n}(\bar{X} - \mu)$ behaves approximately like a normal random variable with finite variance.

Step 3: Implication for $O_p(\cdot)$

Since convergence in distribution to a non-degenerate random variable implies boundedness in probability, we have:

$$\sqrt{n}(\bar{X} - \mu) = O_p(1)$$

Step 4: Rearranging the Expression

Dividing both sides by $\sqrt{n}$:

$$\bar{X} - \mu = O_p\left(\frac{1}{\sqrt{n}}\right)$$

Interpretation:

- The estimation error $\bar{X} - \mu$ shrinks at the rate $1/\sqrt{n}$.
- As the sample size increases, the estimator becomes more precise.
- The factor $\sqrt{n}$ is the natural scaling that stabilizes the distribution of the estimation error.

Intuition:

- Each observation contributes information about μ .
- Averaging reduces variability, but only at the rate $1/\sqrt{n}$.
- Doubling the sample size does *not* halve the error; instead, it reduces it by a factor of $1/\sqrt{2}$.

7.3.5 Slutsky's Theorem

Slutsky's theorem is a central result in asymptotic theory that allows us to combine sequences of random variables that converge in different ways (e.g., in distribution and in probability). It is especially important in deriving the asymptotic distributions of estimators and test statistics.

Statement:

Let $\{X_n\}$ and $\{Y_n\}$ be sequences of random variables such that:

$$X_n \xrightarrow{d} X \quad \text{and} \quad Y_n \xrightarrow{p} c$$

where c is a constant. Then:

- **Addition:**

$$X_n + Y_n \xrightarrow{d} X + c$$

- **Multiplication:**

$$X_n Y_n \xrightarrow{d} cX$$

- **Division (if $c \neq 0$):**

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}$$

Key Idea:

If a sequence Y_n converges in probability to a constant c , then for large samples it behaves approximately like that constant. Therefore, in asymptotic analysis, Y_n can effectively be replaced by c .

Interpretation:

- Convergence in probability to a constant implies that the variability of Y_n vanishes asymptotically.
- As $n \rightarrow \infty$, Y_n becomes increasingly close to c with high probability.
- Hence, combining X_n with Y_n is asymptotically equivalent to combining X_n with c .

Why It Matters:

In practice, many statistics involve unknown parameters (e.g., variance σ^2). Slutsky's theorem allows us to replace these unknown quantities with consistent estimators without affecting the asymptotic distribution.

Example 7.3.2**[Studentized Sample Mean]**

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Define:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

We know:

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

and:

$$S^2 \xrightarrow{p} \sigma^2 \quad \Rightarrow \quad S \xrightarrow{p} \sigma$$

Consider the statistic:

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu)}{S}$$

Applying Slutsky's theorem:

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} \xrightarrow{d} \frac{N(0, \sigma^2)}{\sigma} = N(0, 1)$$

Conclusion:

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

This justifies the use of standard normal approximations in large samples.

Example 7.3.3**(Replacing Unknown Variance)**

Suppose:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, V)$$

but V is unknown and we have a consistent estimator $\hat{V}_n$ such that:

$$\hat{V}_n \xrightarrow{p} V$$

Then:

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\sqrt{\hat{V}_n}} \xrightarrow{d} N(0, 1)$$

Important Notes:

- The sequence Y_n must converge to a **constant**, not a random variable.
- Convergence in probability is stronger than convergence in distribution, which is why the theorem works.
- The theorem is often used together with the Central Limit Theorem.

Connection to Practice:

Slutsky's theorem underpins:

- Construction of t -statistics and z -statistics
- Asymptotic confidence intervals
- Econometric estimators (e.g., OLS, MLE)

Summary:

- If one component converges to a constant, it can be treated as fixed asymptotically.
- This greatly simplifies derivations of limiting distributions.
- It is one of the most frequently used tools in large sample theory.

7.4 Asymptotic Concepts in Hypothesis Testing

In large-sample (asymptotic) inference, hypothesis testing relies on the limiting distributions of test statistics rather than their exact finite-sample distributions. This allows statisticians to conduct inference even when exact distributions are unknown or difficult to derive.

7.4.1 Test Statistics

A **test statistic** is a function of the sample data used to decide whether to reject a null hypothesis H_0 . In large samples, many test statistics have **asymptotic (approximate) distributions**, often normal or chi-square.

General Idea:

If an estimator satisfies:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$$

then we can construct test statistics that are asymptotically standard normal.

Example 7.4.1

(Wald Test)

The Wald test evaluates whether a parameter θ equals a hypothesized value θ_0 .

$$W = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})}$$

Under H_0 :

$$W \xrightarrow{d} N(0, 1)$$

Interpretation:

- Measures how far $\hat{\theta}$ is from θ_0 in standard error units.
- Large values (in absolute terms) indicate evidence against H_0 .

Decision Rule: Reject H_0 if:

$$|W| > z_{\alpha/2}$$

Example 7.4.2

(Wald Test in Linear Regression)

Consider the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, \dots, n$$

where $\varepsilon_i \sim \text{i.i.d. } (0, \sigma^2)$.

Suppose we want to test whether the explanatory variable X has a significant effect on Y :

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$$

Let $\hat{\beta}_1$ be the OLS estimator of β_1 , with standard error $\text{SE}(\hat{\beta}_1)$.

The Wald test statistic is:

$$W = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)}$$

Asymptotic Distribution:

Under H_0 :

$$W \xrightarrow{d} N(0, 1)$$

Interpretation:

- Measures how large the estimated slope $\hat{\beta}_1$ is relative to its variability.
- If $\hat{\beta}_1$ is far from zero (in standard error units), there is evidence against H_0 .

Decision Rule:

Reject H_0 if:

$$|W| > z_{\alpha/2}$$

Numerical Illustration:

Suppose:

$$\hat{\beta}_1 = 2.5, \quad \text{SE}(\hat{\beta}_1) = 1.0$$

Then:

$$W = \frac{2.5}{1.0} = 2.5$$

At significance level $\alpha = 0.05$:

$$z_{0.025} \approx 1.96$$

Since:

$$|2.5| > 1.96$$

we **reject** H_0 and conclude that X has a statistically significant effect on Y .

Conclusion:

The Wald test provides a simple and widely used method for testing hypotheses about regression coefficients using asymptotic normality.

7.4.2 Likelihood Ratio Test (LRT)

The **Likelihood Ratio Test (LRT)** is a fundamental method in statistical inference used to compare two competing models: a **restricted model** under the null hypothesis H_0 , and an **unrestricted model** under the alternative hypothesis H_1 .

Basic Idea:

The LRT evaluates how well the model under H_0 fits the data relative to the best possible model (without restrictions). If imposing the restrictions significantly worsens the fit, then the null hypothesis is rejected.

Likelihood Function:

Let $L(\theta)$ denote the likelihood function based on observed data:

$$L(\theta) = P(\text{data} \mid \theta)$$

- Under H_0 : parameters are restricted to θ_0
- Under H_1 : parameters are freely estimated (MLE $\hat{\theta}$)

Likelihood Ratio:

The likelihood ratio is defined as:

$$\Lambda = \frac{L(\theta_0)}{L(\hat{\theta})}$$

- $0 \leq \Lambda \leq 1$
- Λ close to 1: H_0 fits well
- Λ close to 0: H_0 fits poorly

Likelihood Ratio Test Statistic:

To obtain a convenient test statistic, we take the logarithm and scale:

$$LR = -2 \ln(\Lambda) = -2 \ln \left(\frac{L(\theta_0)}{L(\hat{\theta})} \right)$$

Equivalently:

$$LR = 2 \left[\ln L(\hat{\theta}) - \ln L(\theta_0) \right]$$

Interpretation:

- Measures the **loss of fit** when imposing the restrictions in H_0 .
- Larger values of LR indicate that the unrestricted model fits significantly better.
- If H_0 is true, the difference should be small.

Example 7.4.3

(Likelihood Ratio Test)

Suppose $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$, where σ^2 is known.

We want to test:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0$$

Step 1: Likelihood Function

The likelihood function for μ is:

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(X_i - \mu)^2}{2\sigma^2} \right)$$

Taking logarithms:

$$\ln L(\mu) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

Step 2: Evaluate at MLE

The MLE of μ is:

$$\hat{\mu} = \bar{X}$$

Thus:

$$\ln L(\hat{\mu}) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2$$

Step 3: Evaluate under H_0

$$\ln L(\mu_0) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2$$

Step 4: Compute the Likelihood Ratio Statistic

$$LR = 2 [\ln L(\hat{\mu}) - \ln L(\mu_0)]$$

Substituting:

$$LR = \frac{1}{\sigma^2} \left[\sum_{i=1}^n (X_i - \mu_0)^2 - \sum_{i=1}^n (X_i - \bar{X})^2 \right]$$

Using the identity:

$$\sum (X_i - \mu_0)^2 = \sum (X_i - \bar{X})^2 + n(\bar{X} - \mu_0)^2$$

we obtain:

$$LR = \frac{n(\bar{X} - \mu_0)^2}{\sigma^2}$$

Asymptotic Distribution (Wilks' Theorem):

Under regularity conditions and assuming H_0 is true:

$$LR \xrightarrow{d} \chi_k^2$$

where:

- k = number of restrictions imposed by H_0
- This result is known as **Wilks' Theorem**

Regularity Conditions (Informal):

- The model is correctly specified
- Parameters lie in the interior of the parameter space
- The likelihood function is smooth and differentiable
- The sample size is sufficiently large

Decision Rule:

Reject H_0 if:

$$LR > \chi_{k,\alpha}^2$$

or equivalently, reject if the p -value is less than α .

Graphical Interpretation:

- The unrestricted likelihood $L(\hat{\theta})$ is the maximum possible height of the likelihood function.
- The restricted likelihood $L(\theta_0)$ lies below or equal to it.
- The LRT measures the vertical gap (in log-scale) between these two points.

Example 7.4.4**Numerical Illustration-** from the above example

Suppose:

$$n = 25, \quad \bar{X} = 12, \quad \mu_0 = 10, \quad \sigma^2 = 4$$

Then:

$$LR = \frac{25(12 - 10)^2}{4} = \frac{25 \cdot 4}{4} = 25$$

Step 6: Decision RuleAt significance level $\alpha = 0.05$:

$$\chi_{1,0.05}^2 \approx 3.84$$

Since:

$$25 > 3.84$$

we **reject** H_0 .**Interpretation:**

- The unrestricted model (using $\hat{\mu} = 12$) fits the data much better than the restricted model ($\mu = 10$).
- The large value of LR indicates a substantial loss of fit when imposing H_0 .
- Therefore, there is strong evidence that $\mu \neq 10$.

Key Insight:

This example shows that the likelihood ratio test measures how much the fit deteriorates under the null hypothesis. A large difference in log-likelihoods leads to rejection of H_0 .

Example 7.4.5**(Likelihood Ratio Test in a Regression Model)**

Consider the multiple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i, \quad i = 1, \dots, n$$

where $\varepsilon_i \sim N(0, \sigma^2)$.

We want to test:

$$H_0 : \beta_2 = 0 \quad \text{vs} \quad H_1 : \beta_2 \neq 0$$

Step 1: Define Models

- **Unrestricted model (under H_1):**

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

- **Restricted model (under H_0):**

$$Y_i = \beta_0 + \beta_1 X_{1i} + \varepsilon_i$$

Step 2: Log-Likelihood Function

For the normal regression model:

$$\ln L = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} RSS$$

where $RSS = \sum \hat{\varepsilon}_i^2$ is the residual sum of squares.

Step 3: Evaluate Log-Likelihoods

- Let RSS_u be the residual sum of squares from the unrestricted model
- Let RSS_r be the residual sum of squares from the restricted model

Then:

$$\begin{aligned}\ln L(\hat{\theta}) &= -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{RSS_u}{2\sigma^2} \\ \ln L(\theta_0) &= -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{RSS_r}{2\sigma^2}\end{aligned}$$

Step 4: Compute LRT Statistic

$$LR = 2 \left[\ln L(\hat{\theta}) - \ln L(\theta_0) \right]$$

Substituting:

$$LR = \frac{1}{\sigma^2} (RSS_r - RSS_u)$$

Step 5: Numerical Illustration

Suppose:

$$RSS_r = 120, \quad RSS_u = 100, \quad \sigma^2 = 4$$

Then:

$$LR = \frac{120 - 100}{4} = \frac{20}{4} = 5$$

Step 6: Asymptotic Distribution

Since one restriction is imposed ($\beta_2 = 0$):

$$LR \xrightarrow{d} \chi_1^2$$

Step 7: Decision Rule

At significance level $\alpha = 0.05$:

$$\chi_{1,0.05}^2 \approx 3.84$$

Since:

$$5 > 3.84$$

we **reject** H_0 .

Interpretation:

- The unrestricted model (including X_2) fits the data better than the restricted model.

- The reduction in RSS is substantial enough to conclude that $\beta_2 \neq 0$.
- Therefore, X_2 has a statistically significant effect on Y .

Key Insight:

The LRT in regression compares how much the residual variation decreases when additional variables are included. A large reduction in RSS leads to rejection of the null hypothesis.

7.4.3 Lagrange Multiplier (Score) Test

The **Lagrange Multiplier (LM) Test**, also known as the **Score Test**, is one of the three fundamental tests in asymptotic inference (alongside the Wald and Likelihood Ratio tests). It is used to test whether a set of parameter restrictions holds, by evaluating the behavior of the likelihood function at the null hypothesis.

Basic Idea:

The LM test examines whether the likelihood function would increase if we relaxed the restrictions imposed by H_0 . Instead of estimating the unrestricted model, the test evaluates the **slope (gradient)** of the log-likelihood at the restricted parameter values.

Key Insight:

- If H_0 is true, the likelihood should be maximized (or nearly so) at the restricted parameter values.
- Therefore, the slope (score) of the likelihood function at θ_0 should be close to zero.
- If the slope is large, this suggests that moving away from θ_0 would significantly increase the likelihood, implying that H_0 is false.

Setup:

Let $\ell(\theta) = \ln L(\theta)$ be the log-likelihood function. Suppose we want to test:

$$H_0 : \theta = \theta_0$$

Define:

- **Score vector:**

$$S(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$$

- **Fisher Information matrix:**

$$I(\theta) = -E \left[\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^\top} \right]$$

Test Statistic:

The LM statistic is given by:

$$LM = S(\theta_0)^\top I(\theta_0)^{-1} S(\theta_0)$$

Asymptotic Distribution:

Under regularity conditions and assuming H_0 is true:

$$LM \xrightarrow{d} \chi_k^2$$

where k is the number of restrictions imposed by H_0 .

Interpretation:

- The score vector measures the **direction and steepness** of the likelihood function at θ_0 .
- The information matrix scales this slope by accounting for variability.
- A large LM statistic indicates that the likelihood increases rapidly away from θ_0 , suggesting that H_0 is false.

Geometric Intuition:

- At the true parameter value, the likelihood function should be flat (zero slope).
- The LM test checks whether the slope at θ_0 is significantly different from zero.
- If the slope is steep, the restricted point is not optimal.

Key Advantage:

- Only requires estimation of the model under the null hypothesis.
- Avoids computing the unrestricted estimator, which may be computationally expensive.

Comparison with Other Tests:

- **Wald Test:** evaluates distance from the null using unrestricted estimates
- **LRT:** compares likelihood values of restricted and unrestricted models
- **LM Test:** evaluates slope at the null without estimating the unrestricted model

Asymptotic Equivalence:

$$LM \approx LR \approx W \quad \text{as } n \rightarrow \infty$$

Example (Normal Mean):

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, with σ^2 known.

Test:

$$H_0 : \mu = \mu_0$$

Step 1: Log-Likelihood

$$\ell(\mu) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (X_i - \mu)^2$$

Step 2: Score Function

$$S(\mu) = \frac{\partial \ell(\mu)}{\partial \mu} = \frac{1}{\sigma^2} \sum (X_i - \mu)$$

Evaluated at μ_0 :

$$S(\mu_0) = \frac{n(\bar{X} - \mu_0)}{\sigma^2}$$

Step 3: Fisher Information

$$I(\mu) = \frac{n}{\sigma^2}$$

Step 4: LM Statistic

$$\begin{aligned} LM &= \frac{[n(\bar{X} - \mu_0)/\sigma^2]^2}{n/\sigma^2} = \frac{n(\bar{X} - \mu_0)^2}{\sigma^2} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \\ &= N(0, 1)N(0, 1) = \chi^2 \end{aligned}$$

Result:

$$LM \xrightarrow{d} \chi_1^2$$

7.4.4 Relationship Between Wald, LRT, and LM Tests

- All three tests are asymptotically equivalent:

$$W \approx LR \approx LM \quad \text{as } n \rightarrow \infty$$

- They may differ in small samples but converge to the same result asymptotically.

7.4.5 Asymptotic Size and Power

In hypothesis testing, two fundamental concepts are **size** and **power**. These concepts describe the performance of a statistical test, particularly in large samples where inference is based on asymptotic approximations.

1. Size (Type I Error)

The **size** of a test, also known as the **Type I error probability**, is defined as:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

Interpretation:

- It measures the probability of incorrectly rejecting a true null hypothesis.
- It reflects the **false positive rate** of the test.
- Common choices are $\alpha = 0.10, 0.05, 0.01$.

Asymptotic Size:

In large-sample theory, we are concerned with the limiting behavior of the test. A test is said to have asymptotic size α if:

$$\lim_{n \rightarrow \infty} P(\text{reject } H_0) = \alpha$$

Key Insight:

- The test may not have exact size α in small samples.
- However, as n increases, the rejection probability under H_0 approaches α .
- This justifies the use of asymptotic critical values.

2. Power of a Test

The **power** of a test is defined as:

$$\text{Power} = P(\text{reject } H_0 \mid H_1 \text{ is true})$$

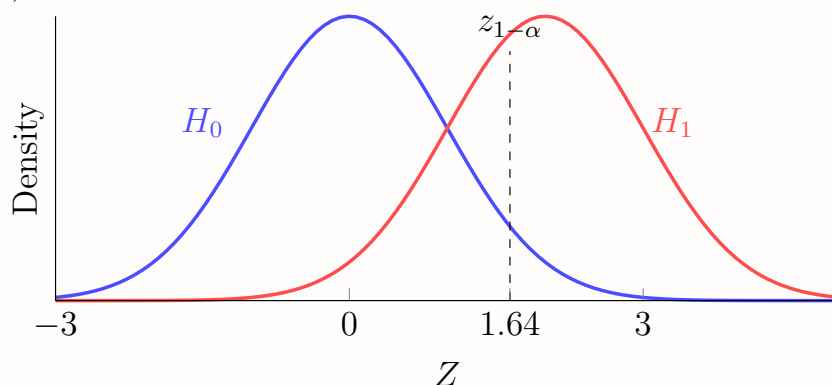
Interpretation:

- It measures the probability of correctly rejecting a false null hypothesis.
- It reflects the **ability of the test to detect an effect**.
- High power means the test is sensitive to deviations from H_0 .

Illustration 7.4.1

Graphical Illustration

Under H_0 , Z follows $N(0, 1)$, while under H_1 , Z follows $N(\delta, 1)$ with $\delta = (\mu_1 - \mu_0)/(\sigma/\sqrt{n})$.



The blue curve (H_0) shows the null distribution, while the red curve (H_1) represents the alternative. The area to the right of $z_{1-\alpha}$ under H_0 is α (Type I error), and the area to the left of $z_{1-\alpha}$ under H_1 is β (Type II error).

Power Function:

The power depends on the true parameter value θ and is often written as:

$$\pi(\theta) = P_{\theta}(\text{reject } H_0)$$

- Under H_0 : $\pi(\theta) = \alpha$
- Under H_1 : $\pi(\theta) > \alpha$

3. Relationship Between Size and Power

- There is typically a trade-off between size and power.
- Reducing α (fewer false positives) often reduces power.
- Increasing α increases power but also increases false rejections.

Goal:

- Maintain size close to the nominal level α
- Maximize power against relevant alternatives

4. Effect of Sample Size on power of the test

We demonstrate the effect of increasing sample size on Power

The rejection region is:

$$|Z| > z_{\alpha/2}$$

Under H_1 , power is:

$$\text{Power} = P(|Z| > z_{\alpha/2} \mid \mu = \mu_1)$$

Since:

$$Z \sim N\left(\frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma}, 1\right)$$

as n increases:

- The mean of Z moves further from 0
- More probability mass lies in the rejection region

Conclusion:

$$\text{Power} \rightarrow 1 \quad \text{as } n \rightarrow \infty$$

Illustration 7.4.2

Let:

$$\mu_0 = 0, \quad \mu_1 = 1, \quad \sigma = 2, \quad \alpha = 0.05$$

Critical value:

$$z_{0.025} = 1.96$$

Case 1: Small sample ($n = 4$)

$$\text{Mean of } Z = \frac{\sqrt{4}(1 - 0)}{2} = 1$$

Case 2: Larger sample ($n = 25$)

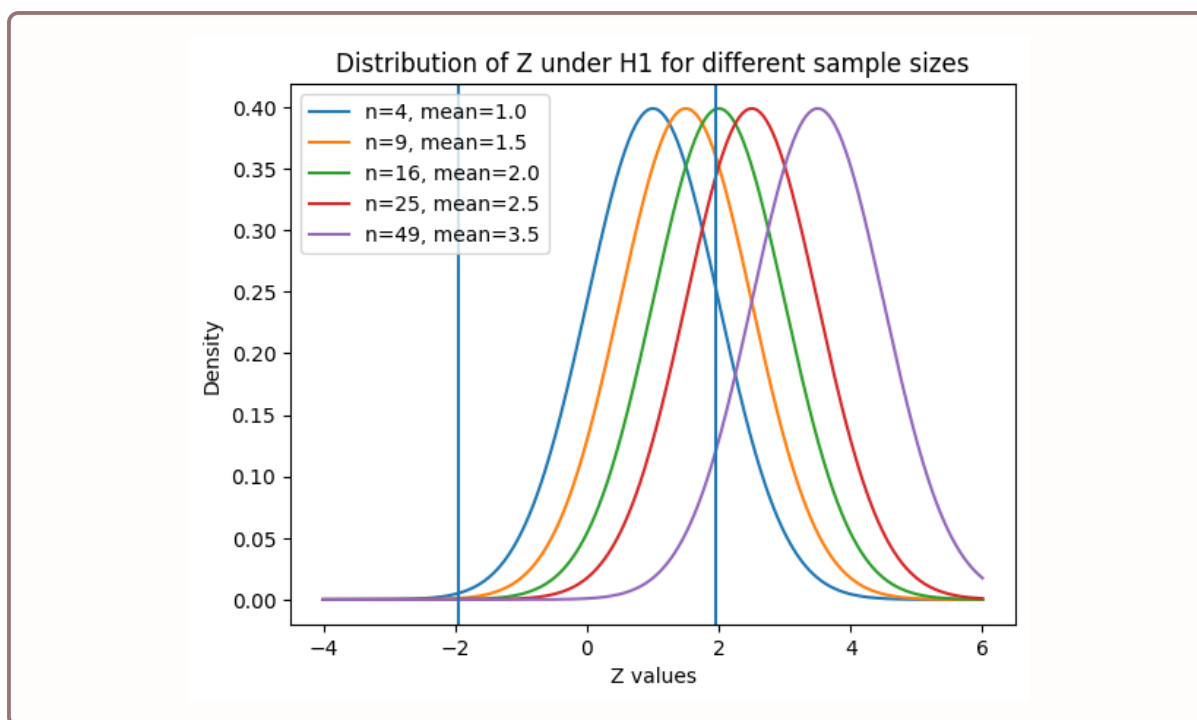
$$\text{Mean of } Z = \frac{\sqrt{25}(1 - 0)}{2} = \frac{5}{2} = 2.5$$

Comparison:

- For $n = 4$: distribution centered at 1 (closer to critical value)
- For $n = 25$: distribution centered at 2.5 (far beyond critical value)

Implication:

- Larger n increases the probability that $|Z| > 1.96$
- Hence, the test is more likely to reject H_0 when it is false.



7.4.6 Consistent Tests

A statistical test is said to be **consistent** if its ability to correctly reject a false null hypothesis improves with sample size, eventually becoming perfect in the limit.

Formal Definition:

A test is consistent if:

$$P(\text{reject } H_0) \rightarrow 1 \quad \text{as } n \rightarrow \infty \quad \text{under } H_1$$

That is, when the alternative hypothesis is true, the probability of rejecting H_0 approaches 1 as the sample size increases.

Interpretation:

- A consistent test will almost surely detect any false null hypothesis given sufficiently large data.
- The test becomes increasingly reliable as n grows.
- Even very small deviations from H_0 can eventually be detected with large enough samples.

Example 7.4.6

(Consistency of a Z-Test)

Let $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$, where σ^2 is known.

We test:

$$H_0 : \mu = 0 \quad \text{vs} \quad H_1 : \mu = 1$$

at significance level $\alpha = 0.05$.

Step 1: Test Statistic

$$Z = \frac{\bar{X} - 0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}\bar{X}}{\sigma}$$

Decision Rule:

Reject H_0 if:

$$|Z| > z_{0.025} = 1.96$$

Step 2: Distribution Under H_1

Under $H_1 : \mu = 1$:

$$\bar{X} \sim N\left(1, \frac{\sigma^2}{n}\right)$$

Thus:

$$Z \sim N\left(\frac{\sqrt{n}(1-0)}{\sigma}, 1\right) = N\left(\frac{\sqrt{n}}{\sigma}, 1\right)$$

Step 3: Power Function

The power is:

$$\text{Power} = P(|Z| > 1.96 \mid H_1)$$

Step 4: Numerical Illustration

Let $\sigma = 2$.

Case 1: Small sample ($n = 4$)

$$\text{Mean of } Z = \frac{\sqrt{4}}{2} = 1$$

$$Z \sim N(1, 1)$$

Power:

$$P(|Z| > 1.96) \approx 0.17$$

Case 2: Moderate sample ($n = 16$)

$$\text{Mean of } Z = \frac{\sqrt{16}}{2} = 2$$

$$Z \sim N(2, 1)$$

Power:

$$P(|Z| > 1.96) \approx 0.52$$

Case 3: Larger sample ($n = 25$)

$$\text{Mean of } Z = \frac{\sqrt{25}}{2} = 2.5$$

$$Z \sim N(2.5, 1)$$

Power:

$$P(|Z| > 1.96) \approx 0.71$$

Case 4: Very large sample ($n = 100$)

$$\text{Mean of } Z = \frac{\sqrt{100}}{2} = 5$$

$$Z \sim N(5, 1)$$

Power:

$$P(|Z| > 1.96) \approx 0.999$$

Step 5: Conclusion

- As n increases, the mean of Z moves further from zero.
- The probability of rejecting H_0 increases.
- Eventually:

$$P(\text{reject } H_0) \rightarrow 1$$

Final Conclusion:

The test is **consistent**, because it will almost surely reject a false null hypothesis as the sample size becomes large.